REFRACTION THROUGH LENS

Time: 1hr **Marks: 40**

Section I

[*All* questions are compulsory]

Question 1: **[5]**

[i] If the intensity of light entering the lens decreases the intensity of image formed by it....

[a] increases

[b] decreases

[c] same size

[d] both a and b

[ii] State the correct condition when the lens has both the focal lengths equal

[a] medium has different refractive index on both sides of lens

[b] medium has same optical densities on either sides of lens

[c] medium has same refractive index on both sides of lens

[d] both b and c

[iii] If rays from a point object after refracting through a lens do not actually meet at a point but they appear to diverge from a point the image is .....

[a] virtual

[b] imaginary

[c] cannot be obtained on screen

[d] all the above

[iv] A convex lens produces an image of the same size as the object when placed at 30cm. Hence radius of aperture of the lens is…

[a] 15cm

[b] 30cm

[c] 60cm

[d] 45cm

[v] Name the lens used in the following instrument to find the defect in the eye.



[a]concave lens

[b] convex lens

[c] plano concave lens

[d]concavo-convex lens

Question 2: **[5]**

[i] An eye specialist prescribes a number of +4.5D to a person for his spectacle lens. Its focal length is …… [10m / 2m / 0.22m]

[ii] An object is placed at 50cm from a converging lens of focal length 20cm. The image formed is real, inverted and … [of the same size / magnified / diminished].

[iii] An object placed at a distance X cm from a converging lens forms a real image at distance of 15cm from lens. If focal length is 10cm the value of X is……. [10cm / 20cm / 30cm]

[iv] In case of a simple microscope the position of object is ……
[between F and 2F / beyond 2F / between F and O] for it to produce a virtual magnified image.

[v] The two types of lenses used in the following apparatus are ……
[convex, convex / concave, concave / convex, concave]



Question 3:

[a] What would be the effect on power of a lens if half the lens is made of tinted glass? Give reason for your answer. [2]

Ans: If half the lens is made of tinted glass there would be no effect on the power of the lens. Only the amount of light passing through reduces the intensity hence the sharpness of image formed decreases. It does not affect the converging or diverging ability of the lens.

[b] The focal length of a convex lens is 0.01mm. Find its power. [2]

Ans: Given focal length of a convex lens = 0.01mm

Hence in cm the focal length would be = 0.001cm = 1×10^{-3}cm

Using the relation,

Power of lens [dioptre] = 100 / focal length in cm

= 100 / [1x 10^{-3}]

= 10^{5} dioptre

[c]Thomas observed that the image of an object is formed at 40cm from the object while the lens is kept exactly midway between object and the image. What will be the power of the lens? [2]

Ans: Since the lens is placed exactly midway between object and image, Object distance is equal to image distance

But given the distance between object and image is 40cm.

Hence object distance is 20cm and image distance is also 20cm.

Using lens formula, 1/f = 1/v – 1/u

= 1/20 – [1/ {-20}]

Hence f = 10cm

Now, power of lens [D] = 100 / focal length in cm

= [100 /10]

= 10D

[d] Can a converging lens behave as a diverging lens. Justify. [2]

Ans: Yes, a converging lens such as a convex lens can behave as a diverging lens if it is placed in a medium having more refractive index than the material of the lens or more optical density than that of lens.

[e] Show by means of a ray diagram how a converging lens can obtain a magnification as one. [2]

Ans: **FIGURE 1**

SECTION II

[Attempt *any two* questions out of three]

Question 4:

[a]

[i] Why does a converging lens usually behave as a burning glass and not a concave lens? [2]

Ans: A converging lens on refraction of parallel beam of rays usually converges at the focus thus increasing the intensity of light rays producing heat hence behaving as a burning glass.

[ii] You are provided with a printed piece of paper. Using this paper how would you differentiate between a convex and a concave lens? [2]

Ans: On placing the lenses on the printed paper if, the letters appear to be magnified and erect we can conclude that the lens is a convex lens [as distance between object and lens is less than focal length] whereas if the print appears to be diminished, we could infer that the lens is a concave lens.

[b] How do you determine the focal length of a lens by using an auxiliary plane mirror? [3]

Ans: Focal length of a lens using an auxillary plane mirror can be obtained as shown in the diagram below:

FIGURE 2

[c] A candle of height 2cm is placed at 0.25m from the eye. Its image is formed on the retina which is 25mm behind the eye lens. Find the size of the image. **[3]**

Ans: Using the relation,

Height of image / height of object = Image distance/ object distance

Height of image = [v/u] x height of object

= [2.5 /{- 25}] x 2

= - 0.2 cm

Hence size of image is – 0.2cm.

Question 5:

[a] Figure below shows four parallel rays of light reaching a thin converging lens. Point F is the focal point of the lens.

FIGURE OF THE QUESTION

[i] A flat screen is placed at F parallel to the broken line.
[ii] The screen is placed beyond F and placed on the broken line.
Describe your observation in both the above cases. [4]

Ans: FIGURE 3 and FIGURE 4

[i] When the flat screen is placed at F parallel to optical axis a bright spot is obtained.

[ii] When the flat screen is placed beyond F, the light would form a blurred spot on the screen.

[b] Draw a ray diagram to illustrate the action of converging lens as a magnifying glass or reading lens. [3]

Ans: FIGURE 5

[c] An object is placed at four times the focal length of a converging lens. Draw a ray diagram to illustrate the same and state the characteristics of image obtained? [3]

Ans: FIGURE 6

The characteristics of image so formed are:

[i] real [ii] inverted and [iii] diminished.

Question 6:

[a] An object is placed at 12cm from a convex lens of focal length 8cm. Without using the formula state the characteristics of image obtained. [3]

Ans: Focal length of convex lens is 8cm so 2[focal length] is 16cm

Object is placed at 12cm from the lens, means between F and 2F.

Hence characteristics of image formed are:

[i] real [ii] inverted and [iii] magnified.

[b] A lens forms an upright and diminished image of an object when an object is placed in front of the lens. Name the lens and draw a ray diagram to illustrate it. [4]

Ans: FIGURE 7

The lens is a concave lens as it is forming an upright and diminished image.

[c] Give any three differences between convex and concave lens? [3]

Ans:

CONVEX LENS	CONCAVE LENS
1. It is usually a converging lens	1. It is usually a diverging lens
2. The image obtained is usually real and inverted except for the case when object is placed between F and O	2. The image obtained is always virtual and erect.
3. Power of this type of lens is positive.	3. The power of this type of lens is negative.